Statistics for the Overall Findings Storyboard (overall_v1)

Colter Bay, Grand Teton National Park, 2022

Nigel K. Anderson · American Museum of Natural History

2026-11-05

All percentages are back-transformed from the log scale: % change in detections per km away from the parking lot, or % difference under white vs. red light. 95% CIs are Wald intervals; p-values are Wald z-tests. "Holm" and "BH" mark p-values adjusted across the five intensities (Holm) or across the taxa tested (Benjamini–Hochberg). Light color and intensity were set experimentally; distance is observational.

Reading and reporting the results

Each figure answers a question at two levels, and they should be reported in that order:

1. **The interaction test**: does the effect differ between conditions (light colors, intensities)? This is one joint Wald χ^2 test per model term, from the same model as the figure.
2. **The effects the figure plots**: the distance slope or white-vs-red difference within each condition, as % change with 95% CI and its (adjusted) p-value.

The asterisks in the figures belong to level 2: each tests whether *that* condition's effect differs from zero. They do not test whether conditions differ from each other. A starred and an unstarred condition are only "different" when the interaction test says so; when it does not, describe the effect as shared across conditions.

Table 1: Interaction tests, 11 stations (Figures 1–2): joint Wald chi-square tests (emmeans::joint_tests) from the models behind the figures.

Figure	Question	Taxon	Chi-sq (df)	p
1B–C / 5B–C	Distance slope, averaged over color and intensity	Bats	4.65 (1)	0.031
		Moths	5.96 (1)	0.015
	Does the distance slope differ between red and white?	Bats	0.04 (1)	0.843
		Moths	9.38 (1)	0.002
2A–B	Does the distance slope differ among intensities?	Bats	7.88 (4)	0.096
		Moths	12.82 (4)	0.012
2C–D	Does the white vs. red difference depend on intensity?	Bats	35.82 (4)	< 0.001
		Moths	24.36 (4)	< 0.001

Table 2: Interaction tests, 13 stations (Figure 5): joint Wald chi-square tests (emmeans::joint_tests) from the models behind the figures.

Figure	Question	Taxon	Chi-sq (df)	p
1B–C / 5B–C	Distance slope, averaged over color and intensity	Bats	0.04 (1)	0.836
		Moths	11.74 (1)	< 0.001
	Does the distance slope differ between red and white?	Bats	0.06 (1)	0.809
		Moths	15.83 (1)	< 0.001

Figure	Question	Taxon	Chi-sq (df)	p
2A–B	Does the distance slope differ among intensities?	Bats	4.40 (4)	0.354
		Moths	15.53 (4)	0.004
2C–D	Does the white vs. red difference depend on intensity?	Bats	39.00 (4)	< 0.001
		Moths	28.99 (4)	< 0.001

Methods paragraph (template)

> Bat detections per site-night and moth specimens per trap night were modeled with negative binomial GLMMs (glmmTMB, nbinom2) including light color × intensity, intensity × distance, color × distance, moon phase, non-forest cover, sky brightness (lights off), day of year (linear and quadratic) and a random intercept for station. Light color and intensity were assigned by night; distance from the lit parking lot is observational. We tested each interaction with a joint Wald χ^2 test and then estimated distance slopes and white-vs-red contrasts within each condition (emmeans), back-transformed to % change per km or % difference. P-values across the five intensities were Holm-adjusted. For the taxonomic hierarchies, zero-inflated negative binomial GLMMs with a taxon × distance interaction were fitted at each level, and p-values were adjusted across taxa with the Benjamini–Hochberg procedure.

Example results, figure by figure

Figure 1B — bats (color × distance not significant, so report one shared slope). Bat detections declined with distance from the lot (-53% per km, 95% CI -76 to -7%, p = 0.031), and the decline did not differ between red and white light (color × distance $\chi^2_1 = 0.04$, p = 0.843). *Avoid:* reporting the red and white slopes as two separate findings.

Figure 1C — moths (color × distance significant, so report each color). The moth distance gradient depended on light color ($\chi^2_1 = 9.38$, p = 0.002): under white light, catches rose away from the lot (+63% per km, 95% CI +27 to +109%, p < 0.001), whereas under red light there was no gradient (+3% per km, 95% CI -21 to +33%, p = 0.843).

Figure 2A — bats (intensity × distance not significant, so describe consistency). The bat decline with distance did not differ significantly among intensities ($\chi^2_4 = 7.88$, p = 0.096); slopes ranged from -42 to -61% per km at every intensity. *Avoid:* "bats declined only at 100%"; the 100% slope is the one individually clear of zero after Holm adjustment, not one that differs from the others.

Figure 2B — moths (intensity × distance significant, so per-intensity results can be interpreted). The moth gradient depended on intensity ($\chi^2_4 = 12.82$, p = 0.012): catches rose away from the lot at 50% (+76% per km, Holm p = 0.011) and 70% (+74% per km, Holm p = 0.003), with no clear gradient at the other intensities.

Figure 2C–D (color × intensity significant for both, so report each intensity's contrast). The white vs. red difference depended on intensity for bats ($\chi^2_4 = 35.82$, p < 0.001) and moths ($\chi^2_4 = 24.36$, p < 0.001). Bats were 34–51% less active under white than red light at 30–70% intensity (all Holm p at most 0.003), with no difference at 10%. Moths showed no color difference at 10–70% (all Holm p at least 0.31). At 100%, the differences reverse for bats (+51%) and are large for moths (+190%); report these with the caveat that the red-light 100% nights include the July full moon and the two wettest nights of the study.

Figures 3–4 (hierarchies): report the count tested, the count significant, and each node's slope with its adjusted p. Of 51 moth taxa tested, 17 changed significantly with distance after Benjamini–Hochberg correction, all increasing away from the lot (e.g. Smerinthinae, +168% per km, adjusted p < 0.001). For bats, give the all-bats slope (single test) and the guild and species slopes BH-adjusted across the 12 nodes (Figure 3 table).

Figure 5 (sensitivity analysis with the far stations). Including the two far stations (0–7.5 km), the bat gradient disappeared (-2% per km, 95% CI -20 to +19%, p = 0.836), while moths still increased away from the lot under white light (+19% per km, 95% CI +12 to +27%, p < 0.001) but not under red (+1% per km, 95% CI -5 to +9%, p = 0.706; color × distance $\chi^2_1 = 15.83$, p < 0.001).

General rules

- Lead with the effect size and its 95% CI; the p-value comes second.
- Never infer that two conditions differ because one is significant and the other is not; cite the interaction test.
- State the adjustment for every family of tests: Holm across the five intensities, BH across hierarchy nodes or taxa, none for single tests.
- Distance is observational ("associated with distance from the lot"); light color and intensity were randomized by night, so causal wording is appropriate for them.

Model formulas

All models are fitted with `glmmTMB` (negative binomial, `nbinom2`). `site` is a random intercept; light `color` (R/W) and `intensity` (10, 30, 50, 70, 100%) are factors; `dist_km` = distance from the parking lot; `jd_c` = day of year, centered.

Figures 1B–C, 2 and 5B–C — one model per taxon (Figures 1–2: 11 stations; Figure 5: 13 stations, GRTE15 excluded). Figure 1 uses the color × distance slopes; Figure 2 uses the intensity × distance slopes (A–B) and the color × intensity contrasts (C–D).

Bats (total detections per site-night; no zero-inflation):

```
total_det ~ color:intensity + intensity:dist_km + color:dist_km + color + intensity
  + dist_km + mean_phase + pct_nonforest + brightness_dark + jd_c + I(jd_c^2)
  + (1 | site)
```

Moths (specimens per trap night; no zero-inflation):

```
detections ~ color:intensity + intensity:dist_km + color:dist_km + color + intensity
  + dist_km + mean_phase + pct_nonforest + jd_c + I(jd_c^2) + brightness_dark
  + (1 | site)
```

Figure 3 — bat hierarchy (zero-inflated, `ziformula = ~1`; zero-filled site-nights). All bats uses total detections; guilds and species use their own detections with `taxon` = guild or species. Every tier converged with the full covariate set.

```
All bats:          total_det ~ dist_km + color + intensity + mean_phase + pct_nonforest
                     + brightness_dark + jd_c + I(jd_c^2) + (1 | site)
Guilds / species:  detections ~ taxon * dist_km + color + intensity + mean_phase
                     + pct_nonforest + brightness_dark + jd_c + I(jd_c^2) + (1 | site)
```

Figure 4 — moth hierarchy (zero-inflated, `ziformula = ~1`; one model per taxonomic level over the taxa with at least 30 site-nights, zero-filled trap nights). If a level's model did not converge, covariates were dropped in this order: non-forest cover, sky brightness, season², moon phase.

```
detections ~ taxon * dist_km + color + intensity + mean_phase + pct_nonforest
  + brightness_dark + jd_c + I(jd_c^2) + (1 | site)
```

In both hierarchies a node's slope and p-value come from the taxon-specific `dist_km` slope (`emtrends`); curves are averaged over color and intensity.

Figure 1A and 5A: Station sky brightness

Table 3: Sky brightness from the fisheye photos (dark sky units). Lights off = darkest lights-off photo (as used in the models); white 100% = mean of three photos (GRTE11 p1 excluded: transient light). Distance from the reference point at GRTE17.

Station	Distance (km)	Lights off (DSU)	White 100% (DSU)	Mapped (DSU)	In Figure 1
GRTE17 (lot)	0.00	2.9	28.6	28.6	yes
GRTE16 (lot)	0.37	15.3	83.2	83.2	yes
GRTE05	0.49	8.4	11.9	11.9	yes
GRTE01 (lot)	0.51	5.9	20.1	20.1	yes
GRTE06	0.77	3.1	3.6	3.6	yes
GRTE02	0.98	4.1	3.5	3.5	yes
GRTE08	1.56	4.3	2.2	2.2	yes
GRTE03	1.60	3.7	3.9	3.9	yes
GRTE09	1.92	4.0	—	4.0*	yes
GRTE11	2.02	4.3	4.9	4.9	yes
GRTE10	2.10	2.7	—	2.7*	yes
GRTE15	7.20	3.3	3.6	3.6	no (far)
GRTE14	7.40	24.0	9.6	9.6	no (far)
GRTE13	7.46	2.8	3.0	3.0	no (far)

* No white-light photos; the lights-off value is mapped instead. GRTE15 appears in neither map. The surface in the maps is inverse-distance-weighted interpolation of log DSU (power 2), clipped to 300 m around the stations.

Figure 1B–C: Distance slopes by light color (11 stations, 0–2.1 km)

Table 4: Models (negative binomial GLMMs; site random intercept).

Model	Obs.	Stations	Family	Converged
Bats: total detections per site-night	502	11	nbinom2	yes
Moths: specimens per trap night	176	11	nbinom2	yes

Both models: color × intensity, intensity × distance, color × distance, moon phase, non-forest cover, sky brightness (lights off), season (day of year, linear + quadratic) and a site random intercept. Full formulas: see *Model formulas* above.

Table 5: Figure 1B — bats: % change in detections per km under each light color (not adjusted).

Light	% per km	95% CI	z	p
Red	-53%	-77 to -7%	-2.16	0.030
White	-53%	-76 to -5%	-2.12	0.034

Table 6: Figure 1C — moths: % change in specimens per km under each light color (not adjusted).

Light	% per km	95% CI	z	p
Red	+3%	-21 to +33%	0.20	0.843
White	+63%	+27 to +109%	3.87	< 0.001

Figure 2: Effects at each light intensity (same models as Figure 1)

Table 7: Figure 2A — bats: % change in detections per km away from the lot, by intensity.

Intensity	Estimate	95% CI	z	p	p (Holm)
10%	-50%	-75 to -0%	-1.97	0.049	0.105
30%	-42%	-72 to +18%	-1.51	0.132	0.132
50%	-53%	-77 to -5%	-2.11	0.035	0.105
70%	-56%	-78 to -11%	-2.30	0.021	0.086
100%	-61%	-80 to -20%	-2.59	0.010	0.048

Table 8: Figure 2B — moths: % change in specimens per km away from the lot, by intensity.

Intensity	Estimate	95% CI	z	p	p (Holm)
10%	-4%	-35 to +41%	-0.21	0.831	1.000
30%	+25%	-16 to +85%	1.11	0.266	0.797
50%	+76%	+21 to +155%	2.98	0.003	0.011
70%	+74%	+27 to +138%	3.45	< 0.001	0.003
100%	-1%	-30 to +39%	-0.08	0.936	1.000

Table 9: Figure 2C — bats: % difference under white vs. red light, by intensity (at the mean distance).

Intensity	Estimate	95% CI	z	p	p (Holm)
10%	+6%	-18 to +36%	0.44	0.657	0.657
30%	-51%	-66 to -31%	-4.03	< 0.001	< 0.001
50%	-40%	-55 to -20%	-3.54	< 0.001	0.002
70%	-34%	-49 to -16%	-3.28	0.001	0.003
100%	+51%	+16 to +97%	3.03	0.002	0.005

Table 10: Figure 2D — moths: % difference under white vs. red light, by intensity (at the mean distance).

Intensity	Estimate	95% CI	z	p	p (Holm)
10%	-3%	-47 to +76%	-0.11	0.909	1.000
30%	-41%	-67 to +6%	-1.76	0.078	0.310
50%	-27%	-55 to +20%	-1.23	0.219	0.658
70%	+5%	-30 to +58%	0.23	0.817	1.000
100%	+190%	+87 to +351%	4.74	< 0.001	< 0.001

Red-light nights at 100% intensity: moth traps 4 nights (12–13 July, 5–6 August 2022), bat detectors 6 nights (11–13 July, 4–6 August); they include the July full moon and the two wettest nights of the study.

Figure 3: Bat distance hierarchy (11 stations)

Table 11: Figure 3 — distance slope at each node. All bats: single test (unadjusted); guilds and species: Benjamini–Hochberg across the 12 nodes.

Tier	Taxon	% per km	95% CI	z	p (raw)	p (adj.)
All bats	All bats	-53%	-76 to -10%	-2.29	0.022	0.022
Guild	Low frequency	-32%	-69 to +48%	-0.97	0.334	0.401
Guild	Myev	-45%	-75 to +21%	-1.49	0.136	0.204
Guild	High frequency	-60%	-81 to -12%	-2.30	0.022	0.065

Tier	Taxon	% per km	95% CI	z	p (raw)	p (adj.)
Species	Epfu	-79%	-91 to -48%	-3.40	< 0.001	0.005
Species	Laci	+15%	-53 to +180%	0.31	0.758	0.758
Species	Lano	+18%	-51 to +187%	0.37	0.712	0.758
Species	Myca	-80%	-92 to -47%	-3.23	0.001	0.005
Species	Myci	-56%	-82 to +9%	-1.77	0.076	0.152
Species	Myev	-45%	-78 to +34%	-1.33	0.185	0.247
Species	Mylu	-60%	-83 to -1%	-1.99	0.046	0.111
Species	Myvo	-78%	-91 to -46%	-3.30	< 0.001	0.005
Species	Myyu	-53%	-81 to +19%	-1.59	0.112	0.192

Zero-inflated negative binomial GLMMs of detections per site-night (all bats = total; guild and species = their detections, zero-filled) with taxon × distance, light color, intensity, moon phase, non-forest cover, sky brightness, season and a site random intercept. Covariates were dropped in order if a model did not converge; covariates used: all bats — mean_phase + pct_nonforest + brightness_dark + jd_c + I(jd_c^2); guilds — mean_phase + pct_nonforest + brightness_dark + jd_c + I(jd_c^2); species — mean_phase + pct_nonforest + brightness_dark + jd_c + I(jd_c^2).

Figure 4: Moth distance hierarchy (11 stations)

Table 12: Figure 4 — the 17 taxa shown: Benjamini–Hochberg-adjusted p < 0.05 across all 51 taxa tested.

Level	Taxon	% per km	p (raw)	p (BH)
Family	Erebidae	+100%	< 0.001	< 0.001
Family	Lasiocampidae	+105%	< 0.001	< 0.001
Family	Sphingidae	+102%	< 0.001	0.001
Family	Notodontidae	+66%	0.014	0.043
Subfamily	Smerinthinae	+168%	< 0.001	< 0.001
Subfamily	Arctiinae	+113%	< 0.001	< 0.001
Subfamily	Sterrhinae	+170%	< 0.001	< 0.001
Subfamily	Lasiocampinae	+108%	< 0.001	0.001
Genus	Smerinthus	+153%	< 0.001	0.002
Genus	Apantesis	+148%	< 0.001	0.002
Genus	Scopula	+149%	0.002	0.009
Genus	Malacosoma	+122%	0.003	0.010
Genus	Feltia	+108%	0.014	0.043
Species	Apamea inficita	+188%	< 0.001	0.004
Species	Smerinthus ophthalmica	+154%	0.001	0.006
Species	Apantesis williamsii	+141%	0.003	0.011
Species	Malacosoma californica	+133%	0.009	0.029

Taxon × distance zero-inflated negative binomial GLMM at each taxonomic level (taxa with at least 30 site-nights), with light color, intensity, moon phase, non-forest cover, sky brightness, season and a site random intercept.

Table 13: All other taxa tested (not significant after adjustment).

Level	Taxon	% per km	p (raw)	p (BH)
Family	Noctuidae	+34%	0.071	0.157
Family	Geometridae	+4%	0.819	0.836
Subfamily	Noctuinae	+37%	0.067	0.157
Subfamily	Sphinginae	-42%	0.114	0.199
Subfamily	Erebinae	+35%	0.175	0.279
Subfamily	Pantheinae	-30%	0.251	0.366
Subfamily	Ennominae	-15%	0.325	0.417

Level	Taxon	% per km	p (raw)	p (BH)
Genus	Euxoa	+81%	0.021	0.060
Genus	Xestia	+111%	0.028	0.076
Genus	Apamea	+58%	0.058	0.148
Genus	Arctia	-48%	0.069	0.157
Genus	Leucania	+59%	0.080	0.169
Genus	Spilosoma	+72%	0.086	0.176
Genus	Trichordestra	-36%	0.098	0.193
Genus	Pero	-37%	0.109	0.199
Genus	Sphinx	-47%	0.112	0.199
Genus	Drasteria	+42%	0.217	0.336
Genus	Panthea	-32%	0.263	0.373
Genus	Agrotis	+29%	0.327	0.417
Genus	Polia	+34%	0.319	0.417
Genus	Macaria	-19%	0.375	0.466
Genus	Egira	+16%	0.518	0.587
Genus	Orthosia	+6%	0.836	0.836
Species	Spilosoma vagans	+71%	0.117	0.199
Species	Sphinx vashti	-47%	0.139	0.229
Species	Trichordestra liquida	-30%	0.243	0.364
Species	Panthea virginarius	-33%	0.302	0.416
Species	Agrotis vancouverensis	+28%	0.457	0.534
Species	Apamea devastator	+30%	0.442	0.534
Species	Egira curialis	+22%	0.461	0.534
Species	Pero behrensaria	-19%	0.562	0.623
Species	Macaria adonis	+11%	0.699	0.758
Species	Leucania commoides	+10%	0.778	0.827
Species	Apamea unanimis	+8%	0.806	0.836

Figure 5B–C: Distance slopes with the far stations (13 stations, 0–7.5 km; GRTE15 excluded)

Table 14: Models (negative binomial GLMMs; site random intercept).

Model	Obs.	Stations	Family	Converged
Bats: total detections per site-night	593	13	nbinom2	yes
Moths: specimens per trap night	210	13	nbinom2	yes

Table 15: Figure 5B — bats: % change in detections per km under each light color (not adjusted).

Light	% per km	95% CI	z	p
Red	-2%	-20 to +20%	-0.18	0.859
White	-2%	-20 to +19%	-0.23	0.816

Table 16: Figure 5C — moths: % change in specimens per km under each light color (not adjusted).

Light	% per km	95% CI	z	p
Red	+1%	-5 to +9%	0.38	0.706
White	+19%	+12 to +27%	5.32	< 0.001

Same model structure as Figure 1B–C, fitted on the 13 stations.